Source-guided msolve diagnostic

Eight fixed local calls retain every outcome for the Boolean ideal over F65537, crossing ordinary/saturation, exact linear algebra1/2 and original/address+undefined builds. Original l1ordinary passes its exact known-answer check; original l2ordinary crashes139 and both saturation calls time out. All four instrumented calls stop earlier at a signed-shift diagnostic. No paid compute. [pcc:unverified]

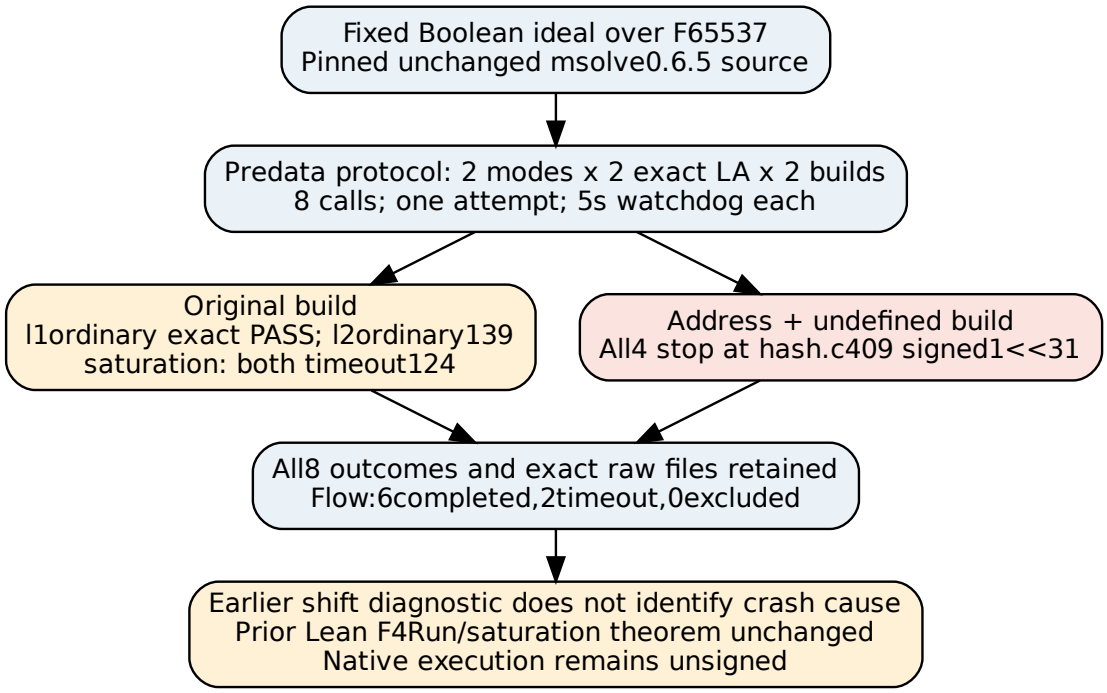

Figure 1: Frozen cells, actual outcomes and scoped interpretation. [pcc:unverified]

Exact outcomes, formal boundary and trust

Build / linear algebra	Ordinary	Saturation
Original /2	exit139, empty output	timeout124 at5s
Original /1	exit0, expected basis passes	timeout124 at5s
Address+undefined /2	exit1, signed-shift diagnostic	exit1, signed-shift diagnostic
Address+undefined /1	exit1, signed-shift diagnostic	exit1, signed-shift diagnostic

Every instrumented stderr identifies neogb/hash.c409, generate_short_divmask: signed1 shifted by31. Source reading shows int32_t res, literal1 and uint32_t return type sdm_t. First-error instrumentation prevents reaching the original crash/timeout sites. This establishes neither their cause nor a complete source fix. All source files remain byte-equal to the official pinned0.6.5tarball. [pcc:unverified]

The original l1ordinary output passes normalized known-basis equality, input reductions, Buchberger S-pairs, staircase3 and three zeros among the four Boolean candidates. This is not exhaustive enumeration of $F65537^2$. A successful fixed output does not prove general msolve conformance to F4Run. The frozen Lean ideal-containment and saturation counterexample retain their original exact polynomial domains. [pcc:unverified]

Flow is8planned,8attempted,6completed,2not_completed(timeouts), 0excluded. Earlier v1planning/dispatch failures and v2ordinary139/ saturation60stimeout remain unchanged. New protocol7bf6fb55 freezes all8cells, exact inputs, analyzer, expected bases and5s per-call/40s batch allowance before data. Source commitf79068d3 precedes affected outputs. No outcome-dependent retry or changed frozen parser. [pcc:unverified]

The export passes all schema, protocol and file-hash checks. Failed metadata drafts/checks are retained separately. Packet PCC-20261110-743f20 remains not_covered: unsigned artifact attestation is not independent execution/recomputation. There is no timing, statistical, generic-system regularity, other-machine or C-refinement claim. [pcc:unverified]

Primary source scope: msolve CLI exact LA1/2 and saturation options, field-width dispatch, F4 linear-algebra paths and reported hash routine; GCC official address/undefined and first-error instrumentation options. KB unavailable, wrapper127; bounded primary/source fallback. Prior formal proof maps remain in the theorem-census and original F4 studies. Next diagnostic: separately preregister address-only instrumentation to observe beyond the earlier undefined-shift check; do not relabel these first-error runs as a memory-error diagnosis. [pcc:unverified]

Exact protocol/source/input/build pins, raw files and all failed outcomes: protocol.json, provenance.json, frozen-files.sha256 and export.json. This report and editable diagram summarize the actual eight-cell record. [pcc:unverified]

Raw argv/stdout/stderr/output/status/timestamps are archived without byte changes in evidence/runs.tar.gz, with hash inventory and roundtrip check. Run `python3 materialize-raw.py` before checking export files in a fresh checkout; existing unequal files are rejected. [pcc:unverified]